Data-driven prediction of full-load electrical power output of a combined cycle power plant using machine learning regression

Technical report

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Abstract

A combined cycle power plant running at full load still varies its net electrical output by about 76 MW as the weather changes. This report predicts that output from four plant measurements (ambient temperature AT, exhaust vacuum V, ambient pressure AP and relative humidity RH), using the public UCI Combined Cycle Power Plant dataset of 9,568 hourly records. The prediction supports two plant decisions: how much output to commit to the grid, and whether the plant is producing what a healthy plant should under the current weather.

An audit found 41 duplicate records, 55 humidity readings above 100% and an exhaust vacuum column whose values exceed ambient pressure in 182 hours, so its stated unit cannot be correct. After removing the duplicates and capping humidity, 9,527 hours were split 80/20, stratified on output, with scaling fitted inside the modelling pipeline. Support vector regression with an RBF kernel, k-nearest neighbours regression and a multilayer perceptron were tuned by 5-fold cross-validation on the training set only.

On the 1,906 held-out hours all three models beat the best result of the reference study (R^2 0.912, MAPE 0.872%). KNN reached RMSE 3.45 MW, MAPE 0.540% and R^2 0.959; SVR and MLP reached RMSE 3.61 and 3.66 MW. KNN was selected because its lead is statistically significant, it over-predicts by more than 5 MW on the fewest hours (5.3%), and each prediction rests on six real past hours. Beyond the training range the SVR predicted rising output in a heatwave, so the deployed model carries an operating-envelope check and a neighbour-density check. Permutation importance and accumulated local effects agree with the thermodynamics: output falls by about 54 MW across the temperature range, the condenser penalty is concentrated between 15 and 30 °C, and humid air lowers output once temperature is fixed. An interactive Tableau Public dashboard presents plant KPIs, model results, a performance alarm and a grid-commitment safety margin.

Keywords: combined cycle power plant, power output prediction, support vector regression, k-nearest neighbours, neural network, condition monitoring

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1. Introduction

1.1 The plant

A combined cycle power plant (CCPP) burns fuel once and generates electricity twice (Figure 1). A gas turbine (Brayton cycle) compresses outside air, burns fuel in it and expands the hot gas through a turbine that drives a generator. Its exhaust leaves at roughly 500 to 600 °C, so a heat recovery steam generator (HRSG) uses it to raise steam, which expands through a steam turbine (Rankine cycle) that drives a second generator. The steam is condensed in a condenser and pumped back to the HRSG.

The plant in this dataset ran at full load in every one of its 9,568 recorded hours, so the operators were not changing the set point. Its net output still moved between 420 and 496 MW. With the throttle fixed, that variation comes from conditions the plant cannot control: the state of the air entering the compressor (temperature, pressure and humidity) and the back pressure in the condenser. Those are the four measured inputs.

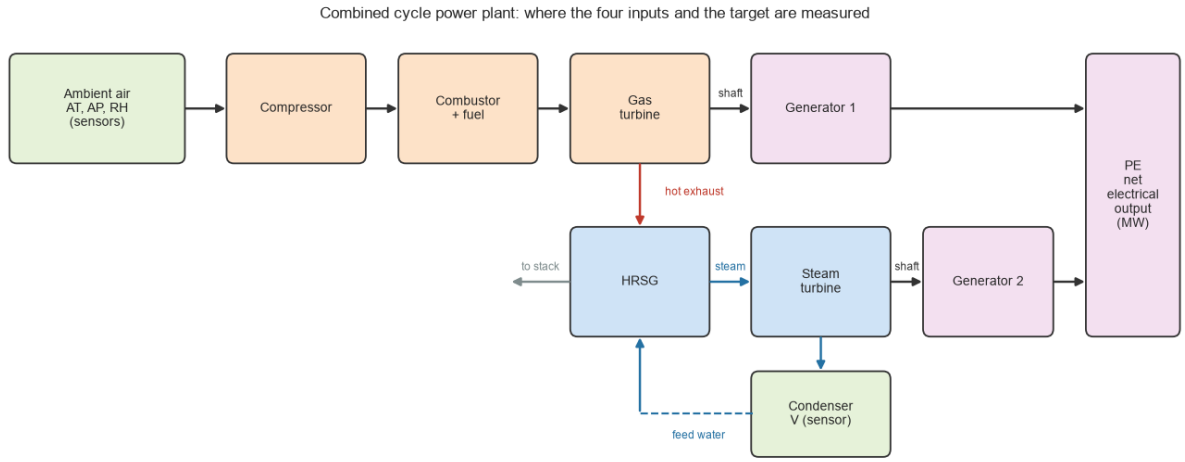


Figure 1. Plant layout and where the four inputs and the target are measured (notebook section 2.1).

1.2 Engineering problem

The problem is to predict the net hourly electrical output PE (MW) of the plant at full load from AT, V, AP and RH. The target is a continuous quantity, so this is a supervised regression task. The prediction supports two decisions:

1. Committing output to the grid. A base-load plant tells the grid operator in advance how many MW it will deliver. Over-predicting means promising energy the plant cannot produce; the shortfall must be covered by other plants and is usually penalized. Under-predicting leaves sellable capacity unused. Over-prediction is usually the more expensive error, and it also affects grid balance.
2. Performance monitoring. At full load, the model gives the output a healthy plant should produce under the current weather. A measured output that stays below the prediction for many hours points to degradation such as compressor fouling, blade wear or condenser fouling. The model is then an expected-performance baseline for condition monitoring.

The two uses need different inputs. For monitoring, all four inputs are live plant readings. For a day-ahead commitment, AT, AP and RH can come from a weather forecast, but V is a plant-side reading that is not known in advance, so section 9.7 also tests a model without V.

1.3 Objectives and success criteria

The targets below were fixed before any model was trained (notebook section 2.5):

1. Accuracy on unseen data: test MAPE below 1% (about 4.5 MW at the mean output), and better than the best model of the reference study by Santarisi and Faouri [3], gradient boosting with R^2 = 0.912 and MAPE = 0.872%.
2. Far better than a trivial model: RMSE below a quarter of the mean-predictor baseline. Always predicting the mean output gives RMSE 17.07 MW but MAPE only 3.26%, because PE varies by a modest share of its mean. MAPE alone therefore flatters weak models, so RMSE and R^2 are reported beside it.
3. Generalization: a gap between training and cross-validated RMSE below 15% of the CV RMSE, and a small spread across folds.
4. Physical plausibility: the learned effects must match the physics of section 2.2 and the data analysis of section 5. Output must fall with AT and V, and the effects of AP and RH must stay small. A model that scores well but breaks these trends, inside or beyond the data, is not deployed.
5. Practical cost: training and prediction time, storage, and how easily the result can be explained to plant engineers.

The reference study used linear, regularized linear and tree-ensemble models and leaned mainly on ambient temperature. This project uses three model families that the study did not test, keeps all four inputs, and checks what the models learned against turbine thermodynamics.

1.4 Scope and assumptions

- The models apply to this plant at full load only. Part-load operation, start-ups and shutdowns are not in the data.
- Inputs and output are hourly averages, so the model predicts hourly average output, not transients.
- Predictions are trusted only inside the observed input ranges. A day hotter than 37.1 °C is an extrapolation.
- The data carries no maintenance records, so slow changes in plant condition over the six years appear as unexplained error.

All numbers in this report are produced by the notebook *CCPP_Power_Prediction.ipynb*. Each section names the notebook section that produced its results, so every value can be traced to code.

2. Dataset

2.1 Source and licence

The data is the Combined Cycle Power Plant dataset of Tüfekci and Kaya in the UCI Machine Learning Repository [2], first analysed by Tüfekci [1]. It was accessed on 24 October 2026 and is published under a Creative Commons Attribution 4.0 (CC BY 4.0) licence, which permits reuse with attribution.

The download is one Excel file, *Folds5x2_pp.xlsx*, with five sheets. It holds 9,568 hourly averages recorded over six years (2006 to 2011) while the plant ran at full load. The file is submitted unchanged in *01_Dataset/raw/*, together with *raw_data.csv*, an exact CSV copy of Sheet1 that the notebook checks value by value against the source.

2.2 Variables

Table 1 lists the variables. Definitions and units come from the dataset authors; ranges are computed from the raw data. The last column gives the direction of effect that thermodynamics predicts, written down before any model was trained so that section 11 can test the models against it.

Table 1. Inputs and target (notebook section 2.2).

Variable	Unit	Acts on	Range	Mean	Expected effect on PE
AT, ambient temperature	°C	gas turbine inlet air	1.81 to 37.11	19.65	strong decrease
V, exhaust vacuum	cm Hg (as labelled)	steam turbine back pressure	25.36 to 81.56	54.31	decrease
AP, ambient pressure	mbar	gas turbine inlet air	992.89 to 1033.30	1013.26	small increase
RH, relative humidity	%	gas turbine inlet air	25.56 to 100.16	73.31	small, sign uncertain
PE, net electrical output (target)	MW	whole plant	420.26 to 495.76	454.37	n/a

The physical role of each input:

- AT. The compressor moves a roughly fixed volume of air per second. By the ideal gas law, ρ = p/(RT), so warmer air means less mass per second, less fuel can be burned at the same turbine inlet temperature, and the gas turbine produces less power. Warm air also raises the compressor work per kilogram.
- AP. Higher pressure gives denser air and more mass flow. Over the observed range (about 4%) the effect should be small.
- RH. Water vapour (18 g/mol) is lighter than dry air (29 g/mol), so humid air is slightly less dense, but vapour also has a higher specific heat. The net sign is not obvious in advance, which makes RH a useful test of what the models learn.
- V. V is measured at the steam turbine exhaust. A lower condenser pressure lets the steam expand further and extract more enthalpy. In this data higher V goes with lower output, so a higher

reading means a higher back pressure. Section 3.2 shows that V cannot be in cm Hg as labelled, so this report treats it as a relative index of back pressure.

- PE. The UCI page calls it "net hourly electrical energy output", which is an hourly average power in MW. The reference study labels it kWh, which does not fit values of 420 to 496.

AT and V are physically linked: on hot days the cooling medium reaching the condenser is warmer, so the back pressure rises. The two inputs are correlated ($r = 0.84$), and their effects have to be separated with care.

2.3 Are the inputs sufficient?

A quick check compares the coldest 5% of hours (mean AT 6.0 °C, PE 485.2 MW) with the hottest 5% (32.2 °C, 432.7 MW). The ideal-gas density of the inlet air, from AT and AP, falls by 9.3% between the two groups, while PE falls by 10.8%. The density change alone accounts for 86% of the output drop. The rest is consistent with higher compressor work and a warmer condenser (mean V rises from about 40 to 71 between the groups). The inputs are physically sufficient to explain the target, which supports a data-driven model built on them.

2.4 Type of learning task and split strategy

The target is continuous, so the task is regression; classifying output as high or low would discard what a grid operator needs. The rows carry no timestamps and the five sheets are shuffled copies, so the data cannot be treated as a time series and a chronological split is impossible. Each row is an independent hourly snapshot at full load, which justifies a randomized train/test split. None of the inputs is calculated from PE, so none leaks the target.

3. Data audit

The audit ran on the raw data and changed nothing. It covers data types, duplicates, missing values, outliers, units, categorical variables, imbalance and leakage. Table 2 lists each finding and the action taken; the notebook saves the same table as *audit_log.csv*.

Table 2. Audit findings and planned actions (notebook section 3.11).

Check	Finding	Action
Data types	all 5 columns float64, 2 decimal places; no text, ID or time columns	none
Categorical variables	none; all inputs are continuous measurements	no encoding needed
Missing values	0 missing cells out of 47,840	no imputation needed
Five Excel sheets	identical rows in five different orders	use Sheet1 only; stacking would copy every row five times
Duplicates	41 exact duplicate rows (0.43%), no conflicting targets	drop the second copy of each
Physical limit: RH	55 rows with RH between 100.06 and 100.16%	cap at 100%, keep the rows
Units: V	182 rows with V above that hour's ambient pressure	keep V; treat as a relative back-pressure index
Outliers	AT, V, PE: none. AP: 88 and RH: 12 by the IQR rule, all plausible weather	keep all
Target coverage	two clusters; 4.4% of hours in the lowest and highest 10 MW bands	stratify the split on PE; report error by band
Multicollinearity	r(AT, V) = 0.84; highest VIF 6.0 (AT)	keep all inputs; read AT and V effects jointly
Leakage	no target-derived, ID or time columns	remove duplicates; scaler inside pipeline; test set used once

3.1 Duplicates and the five sheets

All five sheets of the Excel file hold the same 9,568 rows in different orders. Combining them would repeat every hour five times, and after a random split the same hour would sit in both the training and the test set, which is severe leakage. Only Sheet1 is used.

Within Sheet1, 41 rows are exact copies of another row in all five columns, to 0.01. RH alone has 4,546 distinct values, so two different hours agreeing on all five values by chance is very unlikely, and 41 such matches is not credible. These are recording duplicates. They give a repeated hour double weight in training and, after a random split, can put one copy in each set, which inflates the test score. The reference study did not report them.

3.2 Physical limits and units

RH is the ratio of vapour pressure to saturation pressure and cannot exceed 100%. Fifty-five rows report up to 100.16% (Figure 2, left). They occur on colder-than-average hours (mean AT 15.7 °C against 19.7 °C overall), which fits sensor overshoot on foggy or rainy hours. The other four values in those rows are normal, so deleting the rows would discard valid measurements over an error of 0.16 percentage points.

V is labelled cm Hg, but ambient pressure converted to cm Hg spans only 74.5 to 77.5, and 182 hours have V above that same hour's ambient pressure (Figure 2, right). Neither a condenser vacuum nor a condenser absolute pressure can exceed the surrounding air pressure, so the label is wrong and the true unit cannot be recovered from the data. This does not harm the models: any unit conversion is a linear rescaling, which standardization removes, and the sign of the relationship is unaffected. It does limit interpretation, so V is never quoted as a physical pressure in this report.

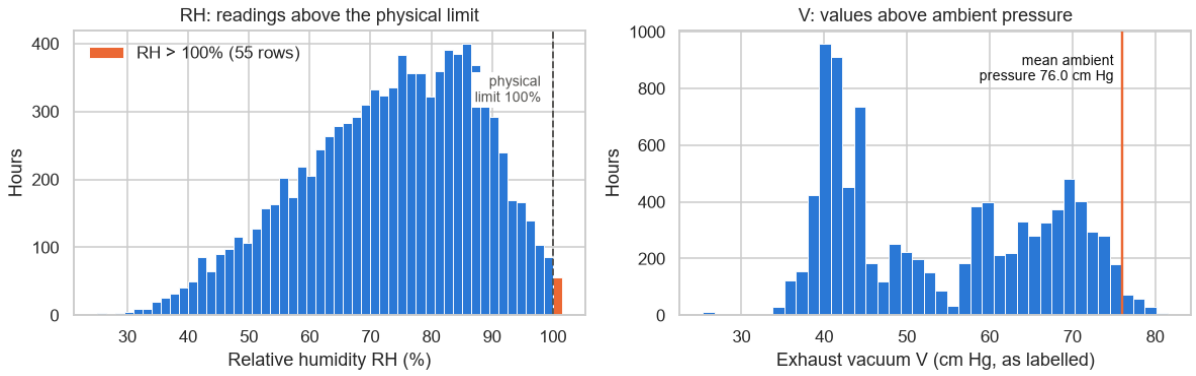


Figure 2. Left: 55 RH readings above the physical limit. Right: V values above the mean ambient pressure, impossible for a vacuum in cm Hg.

3.3 Outliers

The IQR rule and the z-score rule ($|z| > 3$) were applied to every column (Figure 3). AT, V and PE have no outliers by either rule. The 88 flagged AP values (about 993 to 997 and 1030 to 1033 mbar) are ordinary low- and high-pressure weather systems, and the 12 flagged RH values (26 to 31%) are dry, hot hours. In every flagged row the output is inside the normal range. These are real operating conditions, and they are where a model is most likely to be wrong because they are rare. Removing them would improve the test score while making the model worse for exactly the hours an operator cares about. All were kept.

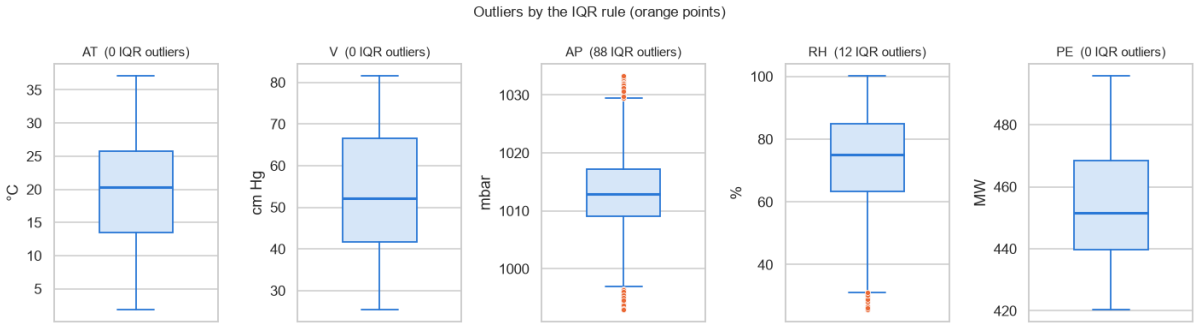


Figure 3. Box plots of the raw variables. Only AP and RH have points beyond the IQR fences, and all are plausible weather.

3.4 Target coverage, correlation and leakage

Class imbalance applies to classification; the regression equivalent is uneven coverage of the target. PE has two clusters, around 430 to 450 and 460 to 480 MW, and the bands below 430 MW and above 490 MW hold only 4.4% of hours between them. The split is therefore stratified on output (section 7.1) and errors are reported per band (section 9.4).

The variance inflation factor is 6.0 for AT, 3.9 for V, 1.5 for AP and 1.7 for RH. This is moderate multicollinearity, above the common warning level of 5 for AT but well below 10. None of the three chosen models needs uncorrelated inputs, so all four are kept, and section 11 reads the AT and V effects with this correlation in mind.

Leakage was checked at every stage. There are no target-derived, ID or time columns. The duplicates and the sheet stacking are the only data-level routes, and both are closed. Scaling is fitted inside a pipeline on training folds only, hyperparameters are tuned by cross-validation on the training set, and the test set is used once, at the end.

4. Preprocessing and feature preparation

4.1 Cleaning rules

Two changes were made to the data, both fixed rules that use no statistic computed from the data: remove exact repeats, and cap RH at its physical limit. Because they learn nothing from the data, applying them before the split cannot leak test information. Every change is logged row by row, so each value in the cleaned file can be traced to the raw file.

Table 3. Cleaning log (notebook section 4.5). The 96 row-level changes are listed in cleaning_changes.csv.

Step	Rule	Rows affected	Rows after
Load raw data	Sheet1 of Folds5x2_pp.xlsx	0	9,568
Remove duplicates	drop exact repeats of all five columns, keep the first	41	9,527
Cap RH	RH above 100% set to 100%	55	9,527
Outliers	kept (real weather, section 3.3)	0	9,527
V unit issue	kept unchanged (section 3.2)	0	9,527

The cleaned data has 9,527 hours. Cleaning moved no column mean by more than 0.03 in that column's unit and no correlation coefficient by more than 0.003. That is the expected outcome: the value of cleaning here lies in correctness, since the test score can no longer be inflated by repeated rows and no input is physically impossible. The notebook asserts after cleaning that no duplicates remain, RH is at most 100%, nothing is missing, and every value not covered by a rule is identical to the raw data.

4.2 What was deliberately not done

- Outliers were not removed or clipped, for the reasons in section 3.3.
- V was not converted. Its true unit cannot be recovered, and standardization makes the unit irrelevant to the models.
- No rows were dropped for being hard to predict. Section 8.5 shows that the hardest hours carry useful information for condition monitoring.
- Scaling was not applied to the saved file. It depends on data statistics, so it is fitted inside the pipeline. The cleaned file keeps every value in its physical unit, which is also what the dashboard needs.

4.3 Scaling and pipeline

The inputs have very different scales: AP varies by a few mbar around 1,013 while V spans more than 50 units. KNN and the RBF kernel of the SVR both work on distances, so an unscaled input with a large spread would dominate them, and the MLP's gradient descent converges slowly and unevenly with unscaled inputs. A StandardScaler (subtract the mean, divide by the standard deviation) is used. A scaler based on the median and IQR was not needed because the outliers are mild.

The scaler is the first step of a scikit-learn Pipeline [4], never fitted on the whole dataset. Each time a pipeline is fitted, on a cross-validation fold or on the full training set, the scaler learns its mean and standard deviation from that training data only and applies them unchanged to the held-out data. For the MLP the target is standardized as well, through a TransformedTargetRegressor that also uses training statistics only (section 8.3).

4.4 A physics-based feature that was tested and rejected

Section 2.3 suggested that AT, AP and RH act on the gas turbine mainly through the density of the inlet air. That density was computed for each hour by treating moist air as a mixture of ideal gases:

$$\rho = p_d/(R_d T) + p_v/(R_v T), \quad p_v = (RH/100)\cdot p_{sat}(T), \quad p_d = p - p_v$$

with R_d = 287.058 J/(kg·K), R_v = 461.495 J/(kg·K), T in kelvin and the Magnus formula p_{sat} = 6.112 exp(17.62 t/(243.12 + t)) hPa for t in °C. The density ranges from 1.121 to 1.298 kg/m³ and correlates with PE at r = +0.947, almost a mirror image of AT (r = −0.984).

Each model type was cross-validated with fixed starting settings on three input sets. Adding density to the four inputs lowered the CV RMSE by 0.038 MW for KNN and 0.014 MW for the MLP and raised it by 0.003 MW for the SVR, all far less than the fold-to-fold spread of about 0.25 MW. The nonlinear models already combine AT, AP and RH internally, so the derived feature adds nothing. It was not used. One result supports the physics: with density and V alone (one variable for each side of the plant) the models lose only 0.35 to 0.6 MW of RMSE against all four inputs.

5. Exploratory analysis and engineering insights

The charts in this section use the cleaned data and each one answers an engineering question (notebook section 5).

5.1 How output responds to each input

Output falls by 2.17 MW per °C of ambient temperature on average, about 0.48% of mean output per degree (Figure 4). Even so, at a fixed temperature PE still scatters by about 5.2 MW (one standard deviation), so temperature alone cannot reach the 1% accuracy target and the other inputs have to explain that scatter. The responses are not straight lines: the PE-AT curve is slightly convex, the PE-V curve falls steeply between about 40 and 50 and flattens above 70, and the PE-AP curve is not even monotonic at the low-pressure end.

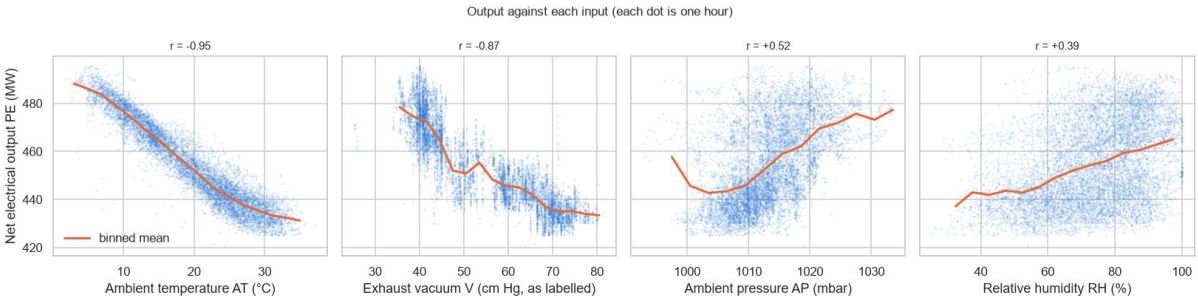


Figure 4. Output against each input, with binned means. r is the Pearson correlation with PE.

5.2 Direct and partial correlation

The Pearson correlation between an input and PE mixes the input's own effect with that of anything it moves with. The partial correlation removes the linear influence of the other three inputs (Figure 5). AT stays strongly negative (Pearson −0.95, partial −0.80) and V keeps a moderate effect of its own (Pearson −0.87, partial −0.31). Two inputs change their story. RH reverses sign, from +0.39 to −0.36, because humid hours are mostly cool hours (r(AT, RH) = −0.54) and cool hours give high output. AP almost vanishes, from +0.52 to +0.07, because high-pressure weather at this site tends to be cold weather.

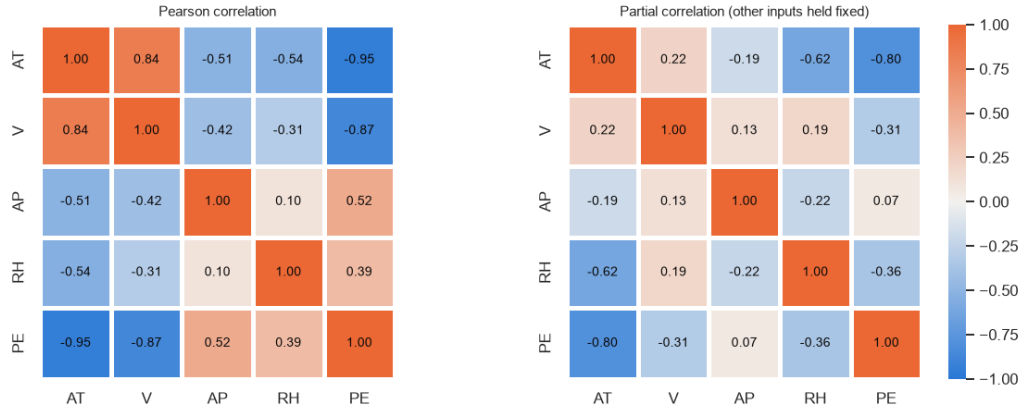


Figure 5. Pearson correlations (left) and partial correlations with the other inputs held fixed (right).

5.3 Temperature and the condenser together

Figure 6 shows two effects at once. V rises with temperature, because the condenser's cooling medium is warmer on hot days. And at a fixed temperature, hours with higher V produce less. That within-band link is weak below 10 °C (r = −0.17) and above 30 °C (r = −0.15), where V hardly varies, and strongest between 15 and 30 °C (r = −0.53 to −0.66), where V spreads widely. The effect of V depends on AT, which is an interaction between inputs.

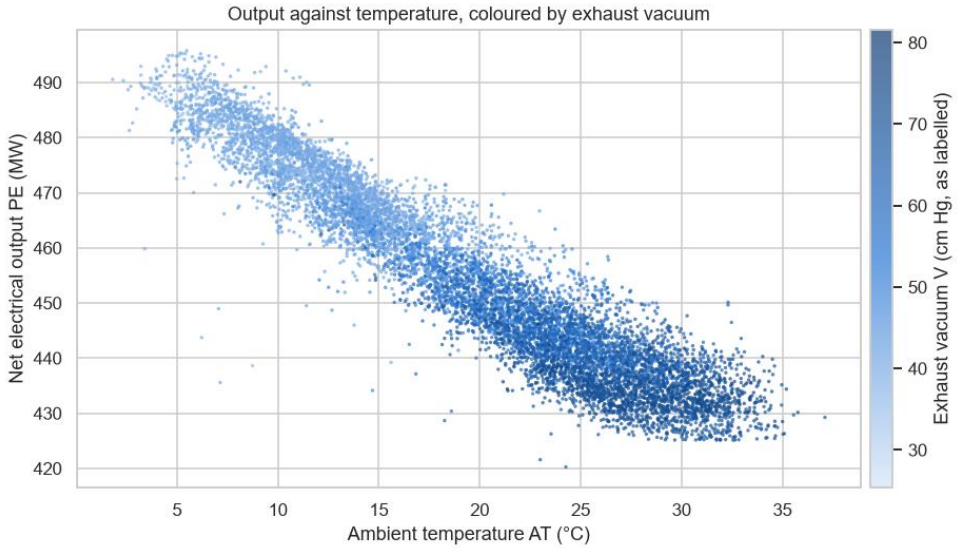


Figure 6. Output against temperature, coloured by exhaust vacuum. At a fixed temperature, darker (higher V) points sit lower.

5.4 What remains after temperature

Subtracting a curved AT trend from PE leaves the output above or below what temperature alone suggests (Figure 7). Against RH this remainder falls steadily from about +4.6 MW on dry hours to about −3.5 MW on saturated hours, a swing of about 8 MW, the clearest secondary effect in the data. Against AP it rises by 3 to 4 MW between 1003 and 1015 mbar and then drifts back, with no consistent direction. Against V it stays within about 2 MW of zero up to about 66 and then dips to nearly −4 MW around 70: the condenser penalty behaves like a threshold.

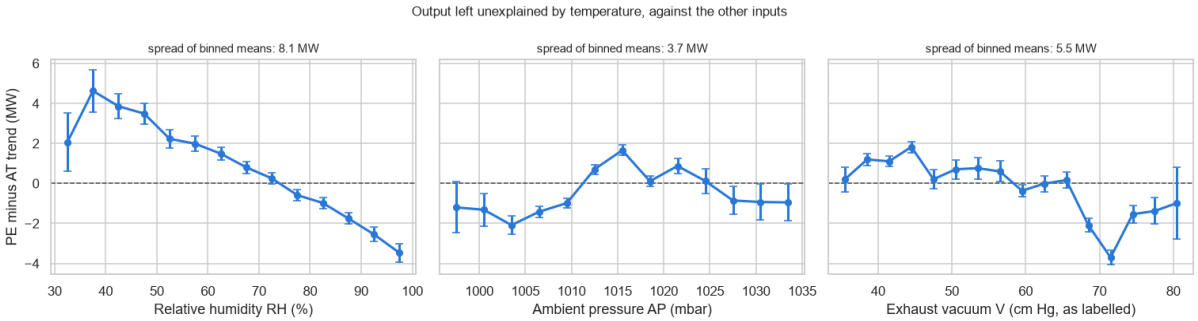


Figure 7. Output left unexplained by a quadratic temperature trend, against RH, AP and V (binned means with 95% intervals).

5.5 Two operating regimes

Splitting the hours at V = 52, the gap between the two V clusters, separates the two output clusters almost cleanly (Figure 8). The low-V regime (4,738 hours) is the cool season: mean AT 14.0 °C and mean PE 467.8 MW. The high-V regime (4,789 hours) is the warm season: mean AT 25.3 °C and mean PE 441.0 MW. The bimodal output found in the audit is therefore a seasonal effect carried through both the gas turbine (AT) and the condenser (V).

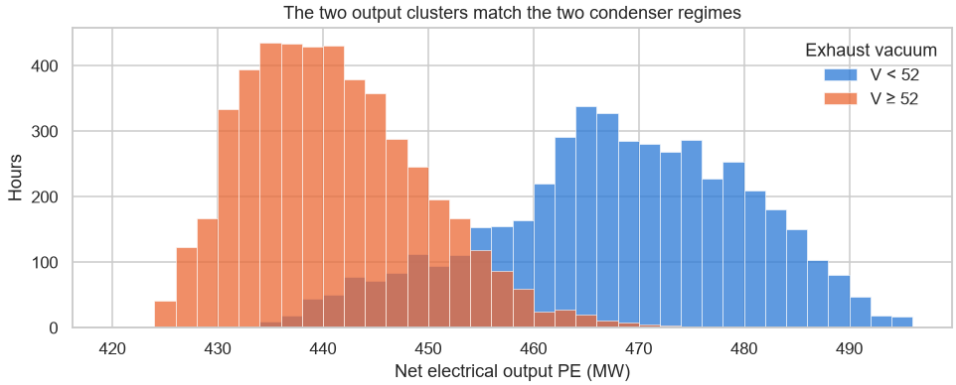


Figure 8. The two output clusters match the two condenser regimes.

5.6 Engineering insights

1. Ambient temperature sets the plant's capacity. Each extra degree costs about 2.2 MW, and the hottest 5% of hours deliver about 52 MW less than the coldest 5%. Most of this follows from the falling density of the compressor inlet air. Sellable capacity on a hot afternoon is far below winter capacity, so commitments should follow the temperature forecast, and inlet-air cooling would act directly on the largest loss.
2. The condenser matters mainly in warm weather. Below about 10 °C V stays near 40 and has little effect. Between 15 and 30 °C V spreads widely, and at the same temperature higher V means lower output, mostly above V ≈ 66. Condenser cleaning and cooling-system maintenance pay off most before and during summer.
3. Humid air lowers output once temperature is accounted for. RH correlates positively with PE only because humid hours are cool hours. At a fixed temperature the effect is negative (partial r = −0.36), worth about 8 MW from dry to saturated air. Reading the plain correlation at face value would have suggested the opposite.
4. Pressure's apparent effect is mostly borrowed from temperature. Its correlation of +0.52 drops to +0.07 with the other inputs fixed, and its own effect is a few MW with no consistent direction. Ranking AP second on correlation alone would be misleading.
5. The relationships are nonlinear and interacting, across two seasonal regimes. The AT response is curved, the V response has a threshold, and the effect of V depends on AT. This is the evidence for choosing nonlinear models that learn interactions over linear ones, and for checking errors separately in each output band.

Sections 9 to 11 test whether the trained models reproduce insights 1 to 4.

6. Interactive dashboard

The dashboard was built in Tableau Public and is published at:

https://public.tableau.com/app/profile/shaikh.muhammad.zain/viz/CCPPfull-loadpowerpredictionoperationsmodelsanddecisions/Decisionsupport

It is built from seven CSV files written by notebook section 13 (*03_Dashboard/data/*), so every number in it matches the notebook. For training hours it shows out-of-fold predictions and for test hours the test predictions, so no error shown was made by a model that had seen that hour. The workbook holds

three dashboards, each 1600 × 1200 pixels, with a title, KPIs, filters or parameters, and a three-line text summary of what the page shows.

6.1 Plant operations

The first dashboard describes the data and the engineering problem (Figure 9). Five KPI tiles show the hours of data (9,527), average output (454.3 MW), highest and lowest output, and average temperature. A scatter of output against temperature, coloured by exhaust vacuum, shows insights 1 and 2. A histogram of output, stacked by condenser regime, shows the two seasonal clusters. A bar chart gives average output by temperature band, from 481.8 MW below 10 °C to 433.0 MW above 30 °C.

Filters for condenser regime, data set (train or test) and an ambient temperature range slider apply to every sheet. Clicking a temperature band filters the whole page to that band: Figure 10 shows the 25 to 30 °C band, where the KPIs change to 2,015 hours and 437.6 MW average output, and the histogram shows that these hours belong almost entirely to the warm-season regime.

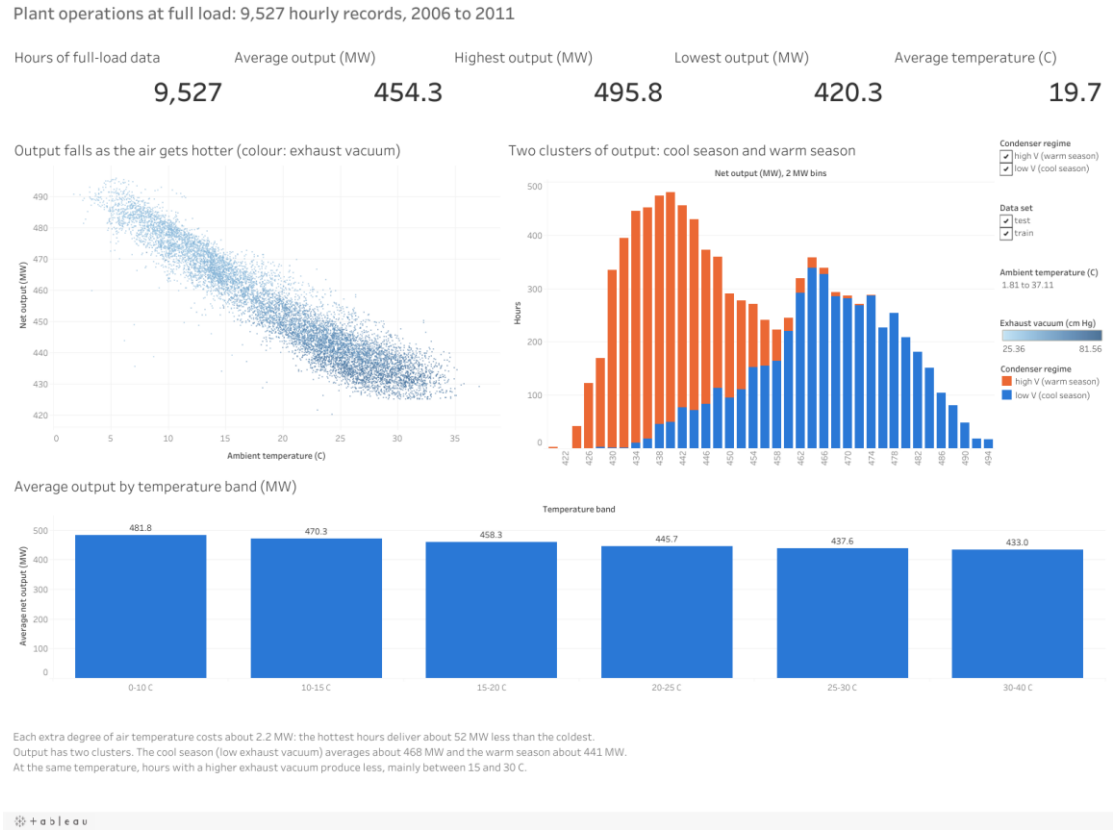


Figure 9. Dashboard 1, Plant operations: KPIs, output against temperature, the two output clusters and output by temperature band.

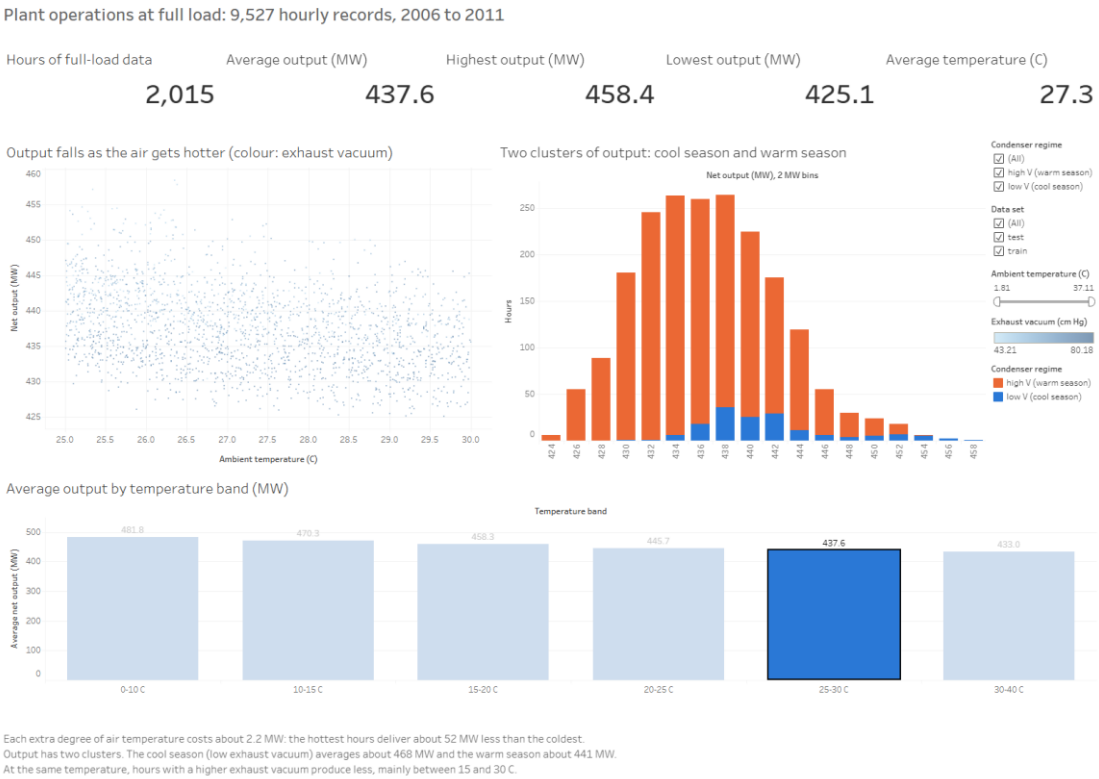


Figure 10. Dashboard 1 after clicking the 25 to 30 °C band. Every sheet and KPI is filtered to that band.

6.2 Model performance

The second dashboard compares the models and explains the selected one (Figure 11). KPI tiles give the KNN test metrics (MAE 2.444 MW, MAPE 0.540%, R^2 0.959, RMSE 3.450 MW). A metric parameter switches the model comparison chart between MAE, RMSE, MAPE and R^2, showing cross-validation and test values for each model, the mean baseline and, for MAPE and R^2, the reference study. A predicted-against-actual plot of the KNN test hours colours alarm hours red. Further sheets show test error by output band, permutation importance, and the accumulated local effect of one input chosen with an input parameter.

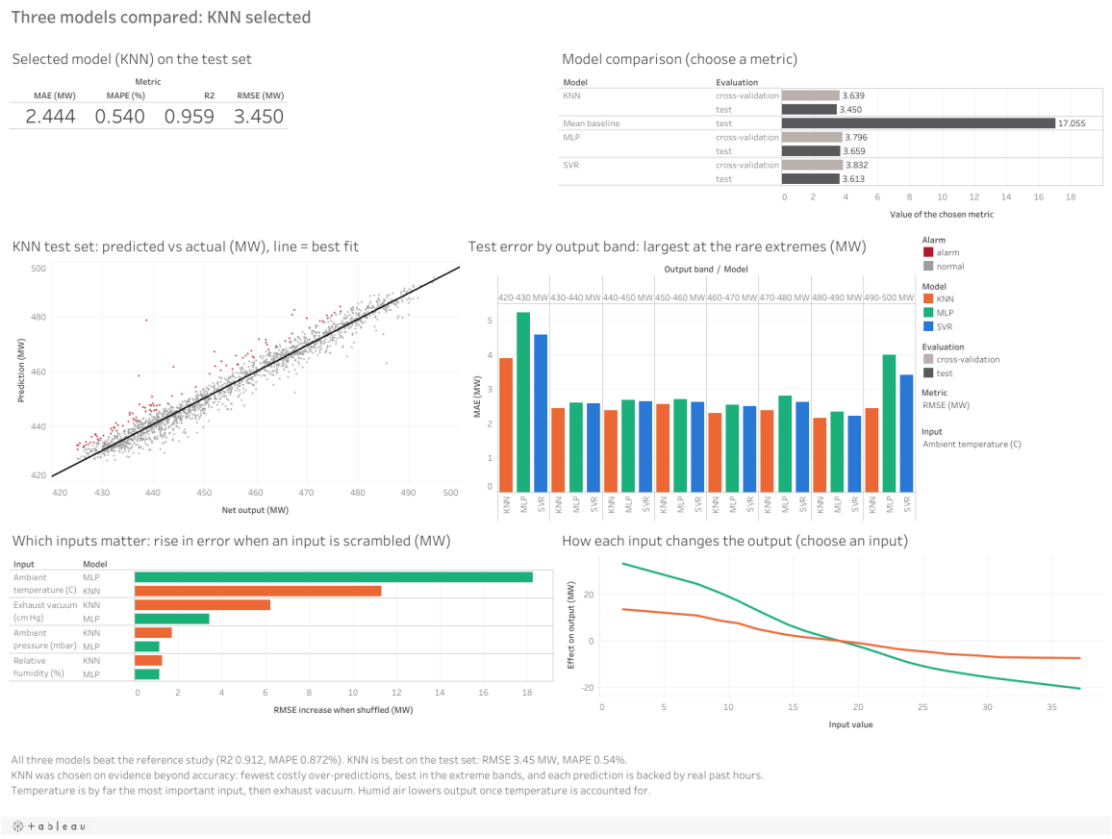


Figure 11. Dashboard 2, Model performance: KNN test KPIs, model comparison, predicted against actual, error by output band, importance and input effects.

6.3 Decision support

The third dashboard turns the model into the two decisions of section 1.2 (Figure 12). An alarm threshold parameter (default 5.2 MW, from section 10.2) sets which hours count as underperformance; the KPI counts them (552 hours at 5.2 MW, 170 at 8 MW), the error scatter marks them in red, and a table lists the worst hours. A safety margin parameter (default 2.5 MW) moves a reference line on two margin charts and updates the share of hours short (19%) and the yearly shortfall and unsold energy (3,658 and 25,219 MWh). A table shows the five unseen cases of section 10 with their status flags.



Figure 12. Dashboard 3, Decision support: performance alarm, safety margin trade-off and the five unseen cases.

Evidence of the demonstration is in *03_Dashboard/*: the packaged workbook (*CCPP_dashboard.twbx*), five screenshots (including the band click and the alarm threshold set to 8 MW), a narrated 2 minute 49 second walkthrough video (*CCPP_dashboard_walkthrough.mp4*).

7. Modelling approach

7.1 Train/test split

Because the data has no timestamps, a randomized split is correct (section 2.4). 80% of the hours (7,621) are used for training and tuning and 20% (1,906) form the test set, with random seed 42. The split is stratified on output deciles so that the rare high and low hours appear in the test set in the same proportion as in the whole dataset. The output bands of the two sets agree to within 0.5 percentage points and their input means to within 0.2 units, and the notebook confirms that no record appears in both. The test set was locked away until the final comparison in section 9 and used once.

7.2 Cross-validation and metrics

Hyperparameters were tuned with 5-fold shuffled cross-validation on the training set. Each fold is held out once while the model is trained on the other four; the mean over folds is the validation score and the spread shows its stability. Four metrics are reported:

- MAE (MW), the typical size of an error.
- RMSE (MW), which weights large errors more. It is the tuning metric, because large errors are the costly ones when committing output.

- MAPE (%), the error relative to output, used for the 1% target.
- R^2, the share of output variation the model explains.

A model that always predicts the mean training output scores a CV RMSE of 17.04 MW and sets the floor every model must beat.

7.3 Choice of algorithms

The data analysis set the requirements: the relationships are smooth but nonlinear, the effect of V depends on AT, there are only four inputs and about 7,600 training hours, and the output must be explainable to plant engineers. Three model families meet these requirements in different ways, and none of them was used in the reference study:

- Support vector regression with an RBF kernel [5] fits a smooth function inside a tube of half-width ε, ignoring errors smaller than ε. The smooth kernel suits the physics of air density and back pressure, ε can be set in MW to roughly match sensor noise, and at this data size training takes seconds. Each of its three hyperparameters (C, γ, ε) has a clear meaning.
- k-nearest neighbours regression predicts the output of a new hour from the k logged hours with the most similar conditions. It makes no assumption about the shape of the relationship, so it can follow the V threshold and the AT-V interaction directly. With 7,600 hours in four dimensions most new hours have close neighbours, and each prediction can be checked against the real hours behind it.
- A multilayer perceptron builds curved responses and interactions from layers of simple units. It is a universal approximator, it is small and fast once trained, and its training cost grows only linearly with the number of rows, so it would scale to years of high-frequency sensor data. It tests whether extra flexibility is worth its cost here.

Each has known weaknesses that the evaluation checks: SVR and MLP can produce values outside the training range, KNN cannot extrapolate and weighs every scaled input equally, and the MLP is stochastic and hard to read directly.

8. Model development

Each model was tuned and evaluated on the training set only, with the same pipeline, folds and metrics (notebook sections 7 to 9). None was adjusted using test results.

8.1 Support vector regression

C and γ were tuned on a 12-point grid with ε fixed at 0.5 MW, after a coarser search had located the optimum between C = 10 and 100 and γ = 0.5 and 4. The error surface has a clear valley (Figure 13): small γ gives a curve too stiff to follow the AT-V interaction, very large γ starts to follow noise, and raising C helps up to a point and then flattens. With C = 30 and γ = 2 fixed, ε was varied from 0.1 to 3 MW. Accuracy hardly changed between 0.1 and 2 MW (CV RMSE 3.856 to 3.832), but the number of support vectors fell from 97% to 52% of the training hours, so ε = 2 MW was chosen. Errors under about 0.4% of output are then treated as noise, a reasonable tolerance for hourly-averaged measurements.

The tuned SVR reached CV RMSE 3.83 MW (fold spread 0.26), MAE 2.80 MW, MAPE 0.618% and R^2 0.949. Its training RMSE of 3.38 MW gives a train-CV gap of 0.46 MW, 12% of the CV error, so it is not overfitting.

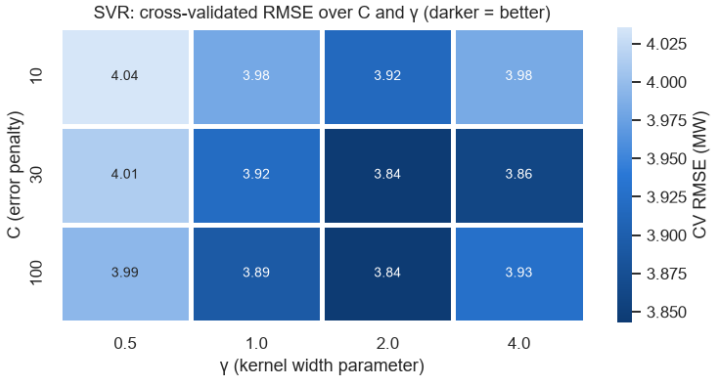


Figure 13. SVR cross-validated RMSE over C and γ (ε = 0.5 MW).

8.2 k-nearest neighbours

Three hyperparameters were tuned together over 136 settings: k from 1 to 30 and then 40, 50, 75 and 100; uniform or inverse-distance weighting; Manhattan or Euclidean distance. The validation curve (Figure 14) shows the bias-variance trade-off directly: k = 1 copies one neighbour's noise, the error falls to a minimum near k = 6, and wider neighbourhoods then mix hours from different conditions. Distance weighting beat equal weighting at every k above 1, and Manhattan distance beat Euclidean, because summing absolute differences lets all four inputs contribute more evenly than squaring them.

The chosen model (k = 6, distance weighting, Manhattan distance) reached CV RMSE 3.64 MW, MAE 2.62 MW, MAPE 0.578% and R^2 0.954. Its training RMSE is exactly zero, which is a property of distance weighting: each training hour finds itself at distance zero and takes all the weight. The honest measures for KNN are the cross-validated error and its fold spread (0.25 MW), which are similar to the SVR's.

Because KNN gives every scaled input equal say in the distance, an uninformative input would only add noise. Removing one input at a time raised the CV RMSE in every case: by 2.40 MW without AT, 0.81 MW without V, 0.42 MW without AP and 0.14 MW without RH. All four inputs carry information, ranked as in section 5.

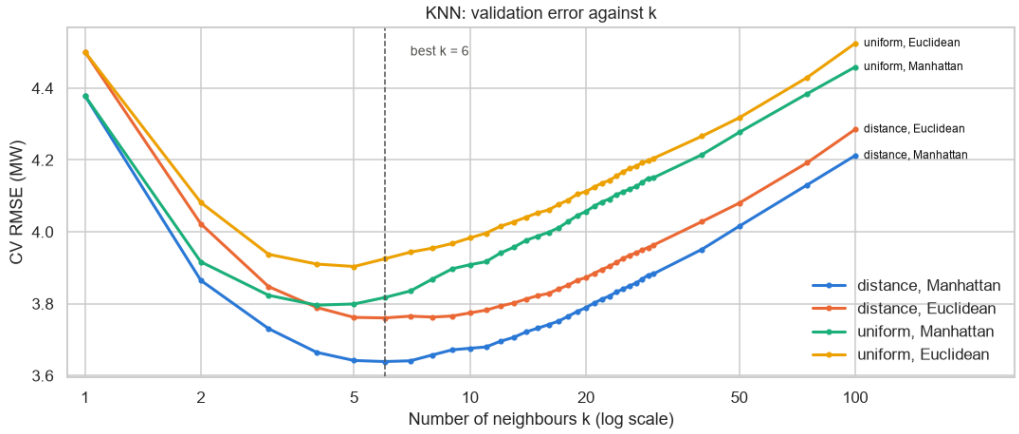


Figure 14. KNN cross-validated RMSE against the number of neighbours, for both weightings and distance measures.

8.3 Multilayer perceptron

Some design choices were fixed before tuning: ReLU activation, the Adam optimizer, a learning rate of 0.001 and early stopping on 10% of each training fold (stop after 30 epochs without improvement). Standardizing the target mattered: PE sits around 450 MW with a spread of only 17 MW, and without scaling the network spends its effort learning that offset. For a (64, 64) network, standardizing the target lowered CV RMSE from 4.18 to 3.94 MW and cut fit time from 24 to 8 seconds.

A 9-point grid over three architectures and three L2 penalties (α) showed that network size mattered more than regularization. The best network, three hidden layers of 128, 128 and 64 units with $\alpha = 10^{-4}$, reached CV RMSE 3.80 MW, MAE 2.81 MW, MAPE 0.618% and R^2 0.950, with a train-CV gap of 0.42 MW (11%). It has 25,473 weights and biases. Across three random seeds the CV RMSE ranged only from 3.796 to 3.815 MW, so the comparison with the other models does not depend on one lucky seed.

The learning curve (Figure 15, right) has training and validation error close together at every size, so the network is not memorizing. The validation error is still falling at the full training size, by about 0.2 MW between half and all of the data, so more operating hours would still improve the MLP a little.

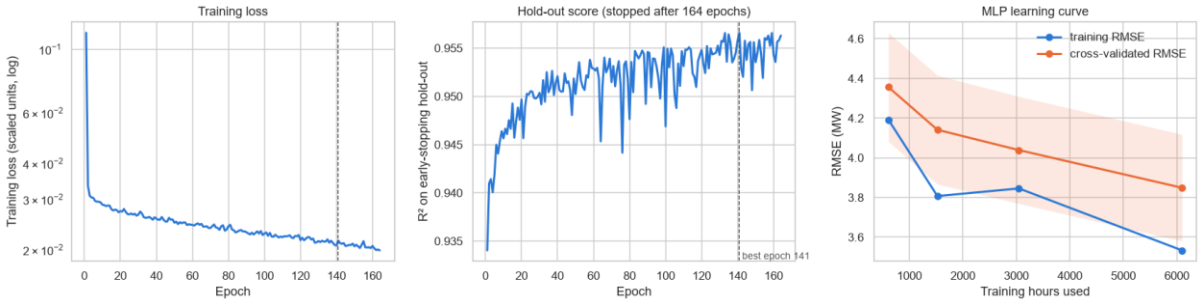


Figure 15. MLP training history (left, middle) and learning curve over training-set size (right).

8.4 Cross-validated comparison

Table 4. Tuned models, 5-fold cross-validation on the 7,621 training hours (mean, with fold standard deviation of RMSE).

Model	Settings	MAE (MW)	RMSE (MW)	MAPE (%)	R^2	Fit time (s)
Mean baseline	always predicts the mean	14.80	17.04 ± 0.08	3.249	−0.001	0.1
SVR	C = 30, γ = 2, ε = 2 MW	2.80	3.83 ± 0.26	0.618	0.949	3.4
KNN	k = 6, distance, Manhattan	2.62	3.64 ± 0.25	0.578	0.954	0.02
MLP	(128, 128, 64), $\alpha = 10^{-4}$	2.81	3.80 ± 0.27	0.618	0.950	25.8

KNN leads by about 0.16 to 0.19 MW. The SVR and MLP differ by 0.04 MW, about the size of the MLP's seed-to-seed variation, so they are tied. The MLP is the most expensive to train; KNN only stores the training hours.

8.5 Where the models fail

Out-of-fold predictions give an honest error for every training hour without touching the test set. All three models show the same pattern (Figure 16 shows KNN). The predictions lie along the diagonal

over the whole range and the error has no trend with temperature. Between 84 and 87% of hours are predicted within ±5 MW and over 98% within ±10 MW.

A small number of hours have far larger errors, almost all over-predictions: the plant produced 24 to 48 MW less than expected. For the SVR, 19 of its 22 errors above 15 MW are over-predictions, and 17 of KNN's 20 such hours are also on the SVR's list. Their inputs are ordinary, many of them cool hours where the plant would normally be near its highest output. Two models built on different principles agree that the weather cannot explain these hours, so something inside the plant must have: a partial trip, a unit below full load for part of the hour, or another unrecorded disturbance. No model using these inputs can predict such hours, and they were deliberately kept. They are real examples of what the condition-monitoring alarm is meant to catch.

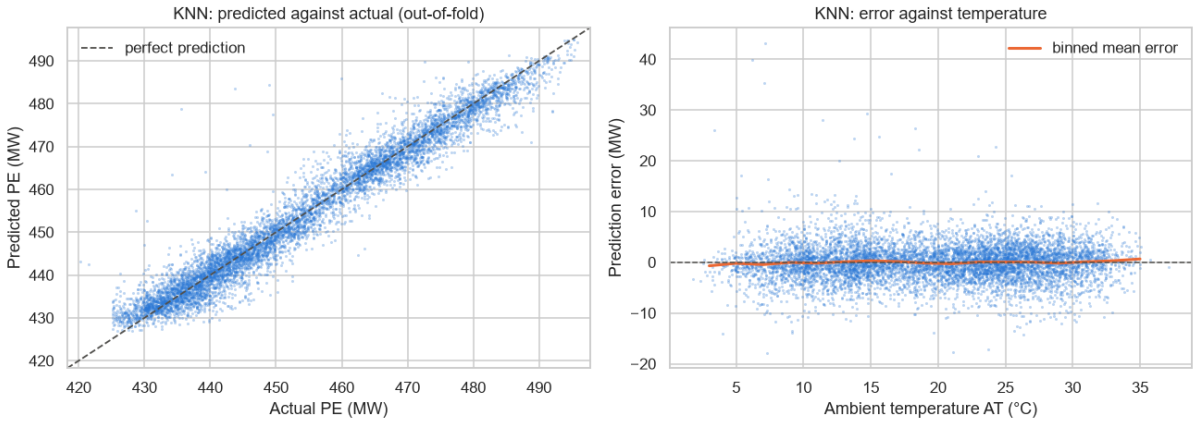


Figure 16. KNN out-of-fold predictions against actual output (left) and error against temperature (right).

9. Model comparison and selection

This is the only place the test set is used. Each tuned model, fitted on the full training set, was applied unchanged to the 1,906 test hours (notebook section 10). The comparison goes beyond one accuracy number: it checks statistical significance, error direction, sensitivity to noise, behaviour beyond the data, a model without V, and cost.

9.1 Test-set accuracy

Table 5. Test-set results on 1,906 unseen hours. The reference row is the best model of Santarisi and Faouri [3], on their own split.

Model	MAE (MW)	RMSE (MW)	MAPE (%)	R^2	CV RMSE (MW)	Test − CV (MW)
KNN	2.444	3.449	0.540	0.959	3.639	−0.190
SVR	2.643	3.613	0.584	0.955	3.832	−0.219
MLP	2.737	3.659	0.604	0.954	3.796	−0.136
Mean baseline	14.826	17.055	3.255	0.000	17.036	+0.019
Reference: gradient boosting [3]	n/a	n/a	0.872	0.912	n/a	n/a

All three models score slightly better on the test set than in cross-validation, within the fold-to-fold spread of about ±0.25 MW, so tuning did not overfit the training data. The ranking is the same as in cross-validation.

9.2 Against the success criteria

Every model meets the accuracy criteria of section 1.3: test MAPE below 1% and RMSE below a quarter of the baseline. Every model also improves on the reference study, with R^2 of 0.954 to 0.959 against 0.912 and MAPE of 0.54 to 0.60% against 0.872%. The two studies used different test splits, so the comparison is indicative, but the margin is large. The overfitting criterion is met by SVR and MLP (gaps of 12% and 11%); it does not apply to KNN, whose test and CV errors agree. Plausibility and cost are assessed below.

9.3 Are the differences real?

The models differ by fractions of a MW, so the test set was resampled with replacement 2,000 times and the RMSE difference of each pair recorded on the same hours (a paired bootstrap). KNN is better than SVR by 0.17 MW (95% interval 0.04 to 0.28) and better than the MLP by 0.21 MW (0.09 to 0.32); neither interval includes zero. The MLP minus SVR difference, +0.05 MW (−0.01 to +0.11), is not significant. KNN's lead is real, but at about 0.04% of mean output it is small next to the 2.5 MW typical error, so accuracy alone does not decide the choice.

9.4 Errors across the output range and their direction

Errors are smallest in the well-populated middle bands and largest at the rare extremes (Figure 17). At the low end the models over-predict and at the high end they under-predict, because predictions are pulled towards the dense middle of the data. The models are therefore slightly optimistic in the hottest, lowest-output hours, which is the direction that risks over-committing. KNN has the smallest error in both extreme bands (3.89 and 2.44 MW MAE, against 5.23 and 3.99 MW for the MLP).

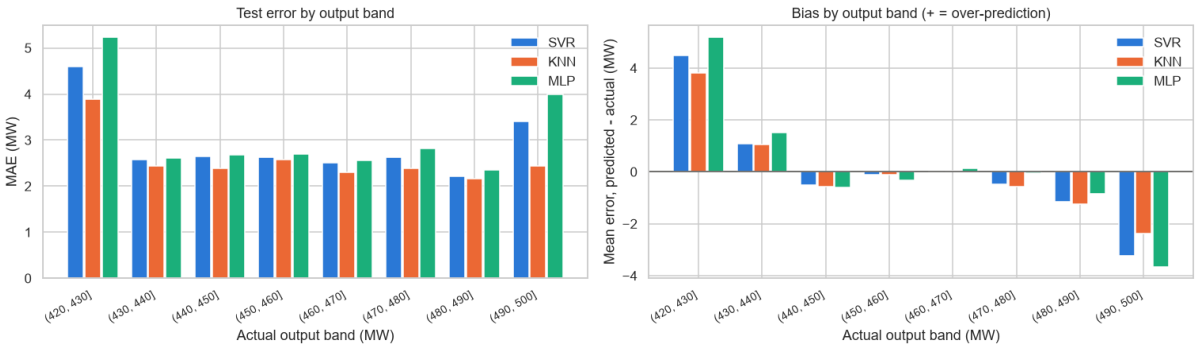


Figure 17. Test MAE (left) and mean signed error (right) by actual output band. Positive error is over-prediction.

For grid commitment, over-predictions above 5 MW are the costly errors. Table 6 shows that KNN makes them on the fewest hours.

Table 6. Direction and size of large test errors.

Model	Over-predicted by > 5 MW (% of hours)	Under-predicted by > 5 MW (% of hours)	Worst over-prediction (MW)	95th percentile of |error| (MW)
KNN	5.30	5.88	39.89	7.00
SVR	6.87	5.88	41.51	7.12
MLP	7.76	6.66	41.61	6.90

9.5 Sensitivity to sensor noise

In operation sensors drift, and for day-ahead use AT, AP and RH come from a forecast. Gaussian noise was added to the test inputs at a reference level (×1: 0.5 °C on AT, 1 unit on V, 1 mbar on AP, 2 points on RH) and at two and three times that level (Figure 18). At ×1 the three models are level (RMSE 4.0 to 4.1 MW). At higher noise the MLP degrades least (5.17 MW at ×3) and the SVR most (5.79 MW). KNN's advantage comes from matching new hours to very similar logged hours, and noisy inputs pick the wrong neighbours; the MLP's smooth function turns small input errors into small output errors.

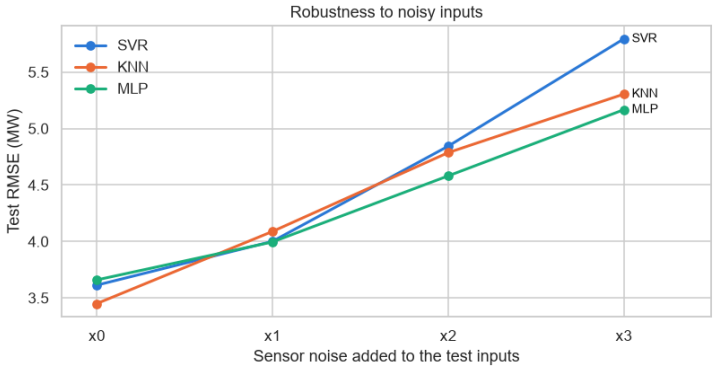


Figure 18. Test RMSE as Gaussian noise is added to the inputs.

9.6 Behaviour beyond the training range

The hottest logged hour is 37.1 °C, and a heatwave can exceed this, which is exactly when capacity is lowest and a good forecast matters most. The models were asked to predict output for AT from −6 to 45 °C, with the other inputs held at the mean values of the coldest or hottest 5% of hours (Figure 19). Physics says output should keep falling with heat and rising with cold.

- SVR gets the direction wrong on both sides. Above about 35 °C its prediction turns upwards, to 455 MW at 45 °C, roughly a mild day's output. Far from all training points the RBF kernel terms fade and the prediction drifts back to the average. In a heatwave it would tell the operator to promise far more than the plant can deliver.
- KNN stays flat, repeating the most extreme logged hours (430.5 MW on the hot side). It cannot be badly wrong in direction, but it does not follow the further loss, so it slightly over-predicts in a heatwave.

- The MLP follows the physical direction on both sides (about −1.5 MW per °C beyond 37 °C), but on the cold side it predicts over 500 MW, above anything the plant has produced. Its trend is plausible, but its values outside the data are unverified.

No model should be trusted outside its training data. Whichever model is chosen needs an operating-envelope check.

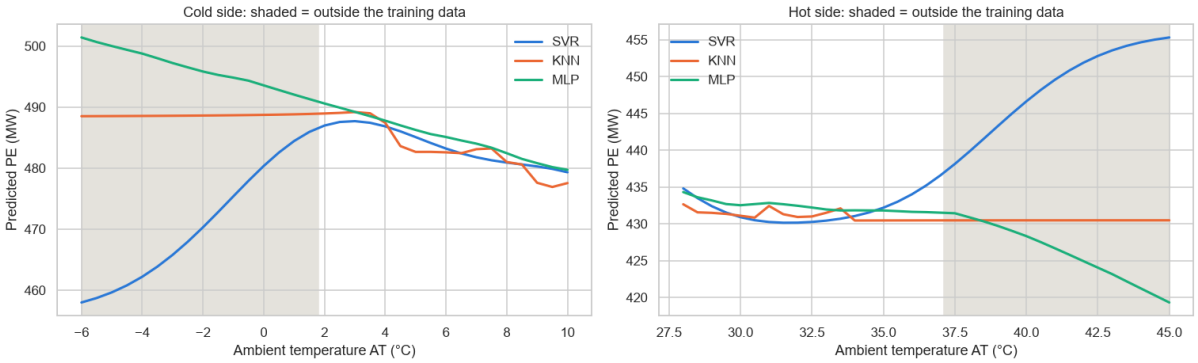


Figure 19. Predictions below and above the observed temperature range. Shaded: outside the training data.

9.7 A day-ahead model without V

Each model was refitted on AT, AP and RH only, with the same hyperparameters (Table 7). The error rises by 0.6 to 0.85 MW, but every model still meets the 1% MAPE target and still beats the reference study. The extra error is the part of the condenser's behaviour that the weather does not determine, such as cooling-water conditions and condenser cleanliness. A day-ahead commitment from the weather forecast is feasible.

Table 7. Test results with and without the exhaust vacuum input.

Model	RMSE, all four (MW)	RMSE without V (MW)	MAPE without V (%)	R^2 without V	RMSE increase (MW)
KNN	3.449	4.294	0.703	0.937	0.845
SVR	3.613	4.235	0.715	0.938	0.622
MLP	3.659	4.326	0.736	0.936	0.666

9.8 Cost and practicality

Table 8. Cost of each model on this hardware (8-core laptop).

Model	Fit time on training set (s)	Prediction time per 1,000 hours (ms)	Saved model size (KB)	Needs training data at prediction time
KNN	0.016	10.8	383	yes
SVR	3.4	392	172	no
MLP	25.8	7.2	610	no

All three are cheap enough for hourly use. KNN trains instantly but must store and search the training hours, so its size and prediction time grow with the data. The MLP's 25,473 weights are about 200 KB; the saved file is larger because scikit-learn also keeps the optimizer state.

9.9 Selection

Table 9. Evidence for the final choice.

Criterion	SVR	KNN	MLP
Accuracy on clean test data	tied second	best, by 0.17 to 0.21 MW (significant)	tied second
Accuracy in the extreme output bands	middle	best	worst
Costly over-predictions (> 5 MW)	6.9% of hours	5.3% of hours (fewest)	7.8% of hours (most)
Noisy inputs (×3)	worst	middle	best
Behaviour beyond the data	wrong direction on both sides	flat: repeats the most extreme logged hours	physical direction, unverified values
Training cost	seconds	instant	tens of seconds
Direct interpretability	low	medium: each prediction rests on real past hours	low

Selected model: KNN regression (k = 6, distance-weighted, Manhattan distance), deployed with an operating-envelope check. The reasons, in order of weight for the two uses in section 1.2:

1. It makes the fewest costly errors. KNN over-predicts by more than 5 MW on the fewest hours and has the smallest bias in the hottest, lowest-output hours, where capacity is tight.
2. It is the most accurate, and the lead is statistically significant. The lead is small, so it supports the choice without deciding it alone.
3. Its predictions can be explained to plant staff. Each prediction is a weighted average of six real past hours, which can be listed. When measured output falls short, an engineer can see which past hours the plant is being compared with.
4. It updates without retraining. New operating hours can be added directly, so its coverage grows as the plant logs more data, including hotter or colder hours than any seen so far.

Its weaknesses are handled. It cannot extrapolate, but neither can the others be trusted there, and the SVR is actively misleading; the deployed model flags any hour whose inputs fall outside the training range (AT 1.81 to 37.11 °C, V 25.36 to 81.56, AP 992.89 to 1033.30 mbar, RH 25.89 to 100%). Its lead disappears with noisy inputs, but at realistic noise it is level with the others. The MLP is the runner-up: it is the least sensitive to noise and the only model whose trend beyond the data follows physics, so it is the better choice if inputs are poor and the reference to check KNN against when the envelope check fires. The SVR is not recommended, because its heatwave behaviour is wrong in exactly the conditions where a wrong forecast costs most.

10. Final model on unseen input cases

Five new operating conditions were written for this purpose, and the notebook confirms that none appears in the dataset (notebook section 11). Each tests something specific: a hot summer afternoon, a cold winter night, a mild day once humid and once dry with everything else identical, and a heatwave beyond the data.

Each prediction comes with a 90% error band taken from the final model's test errors: 90% of test hours had an actual output within 5.2 MW of the prediction. Each case also gets two reliability checks. The range check flags inputs outside the training range. The density check flags unusual combinations: if the mean distance to the six neighbours exceeds 0.551 scaled units (the 95th percentile for test hours), the case is marked "few similar past hours".

Table 10. Final KNN model on five unseen cases, with SVR and MLP for comparison.

Case	AT (°C)	V	AP (mbar)	RH (%)	KNN (MW)	90% band (MW)	SVR (MW)	MLP (MW)	Status
1 Hot summer afternoon	33	71	1008	45	433.9	428.7 to 439.1	435.1	434.5	valid
2 Cold winter night	6	40	1020	85	487.9	482.7 to 493.1	488.2	486.9	valid
3 Mild day, humid	20	52	1013	90	453.3	448.1 to 458.5	454.7	453.1	valid
4 Mild day, dry	20	52	1013	45	452.6	447.4 to 457.8	455.6	456.6	caution: few similar past hours
5 Heatwave beyond the data	41	74	1006	30	429.3	424.1 to 434.5 (not valid here)	449.3	430.0	flagged: AT outside training range

10.1 Plausibility of each prediction

The hot afternoon is predicted at 433.9 MW. Its six neighbours are logged hours within about 1 °C, 1.5 units of V, 1.5 mbar and 3% RH of the case, closer than for a typical test hour, with outputs of 427.0 to 437.8 MW. The cold night is predicted at 487.9 MW, 54 MW above the hot afternoon, which matches the seasonal swing of section 5.6. All three models agree within 1.5 MW on both cases.

Cases 3 and 4 expose a real limitation of KNN. The data analysis of section 5.4 says the dry day should produce more than the humid day (about 5 MW over this RH range at a fixed temperature). KNN predicts the two almost equal, with the humid day 0.7 MW higher. The neighbour tables explain why. The humid case has close neighbours, all within about 1.4 °C of 20 °C. Dry hours at 20 °C are rare, so four of the dry case's six neighbours are at 22.7 to 23.8 °C, and their extra warmth costs more output than the dry air gains. KNN weighs every scaled input equally, so it cannot hold temperature fixed while it searches. The MLP gets the direction right (3.5 MW in favour of the dry day), as does the SVR with a smaller difference. Two things follow: the two KNN predictions are still within each other's error band, so the practical forecast error is small, and the density check catches the problem by marking the dry case "caution". KNN should not be used to read off single-input effects, which is why section 11 uses the MLP for that.

The heatwave case is flagged by the range check, and its error band does not apply because it was measured on hours inside the data. KNN repeats the hottest logged hours (429.3 MW) and the MLP gives a similar 430.0 MW. The SVR predicts 449.3 MW, more than for the 33 °C afternoon, which is physically impossible. Without the flag an operator could take the SVR value and promise 15 to 20 MW more than the plant can deliver on the hottest day of the year.

10.2 Condition-monitoring demonstration

The alarm threshold is set from the test errors: on 95% of normal test hours the plant produced no more than 5.2 MW below the prediction. For the cold night (case 2), a normal reading of 485.9 MW gives a 2.0 MW shortfall and no alarm. A degraded reading of 472.9 MW, 15 MW low as a fouled

compressor or condenser problem might cause, exceeds the threshold and raises an alarm. In operation one hour above the threshold would be noted and a run of consecutive hours would raise the alarm. The large over-predictions of section 8.5 are real examples: hours in which the plant produced 24 to 48 MW less than the weather allowed.

11. Engineering interpretation

This section asks what the models have learned and whether it makes engineering sense (notebook section 12). Section 10 showed that KNN cannot isolate one input's effect, so feature effects are read from the MLP, which learned a smooth function, with KNN shown alongside. Predictions still come from the selected KNN model.

11.1 Which inputs matter

Permutation importance [6] shuffles one input across the test hours, breaking its link with the output, and measures the rise in RMSE (Figure 20). Both models give the same picture: AT far ahead (11.3 MW for KNN, 18.3 MW for the MLP), then V (6.2 and 3.5 MW), with AP and RH small (1.1 to 1.7 MW). This matches the physics and the drop-one test of section 8.2. KNN leans more on V because in its distance V counts as much as AT. Because AT and V are correlated, shuffling one creates some unrealistic hours, which can overstate both.

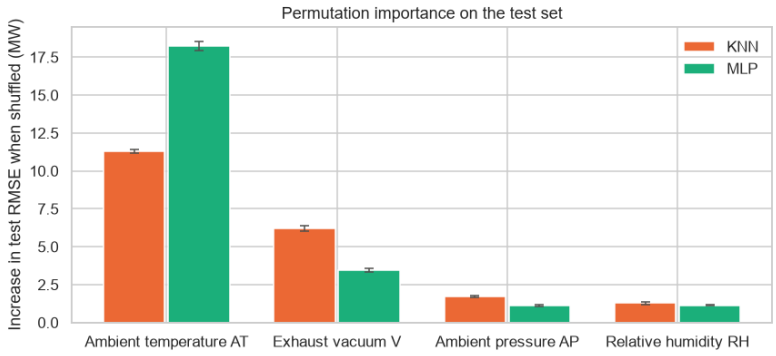


Figure 20. Permutation importance on the test set (rise in RMSE when an input is shuffled, 10 repeats).

11.2 How each input changes the output

A partial dependence plot would set one input to each value for every hour, creating unrealistic combinations such as 35 °C with V = 40. Accumulated local effects (ALE) [7] avoid this: the input's range is cut into 20 bins of equal count, and only hours that really lie in a bin are moved across it. The local differences are summed and centred, giving the effect of the input in MW (Figure 21). Table 11 compares the MLP's learned effects with the expectations of section 2.2.

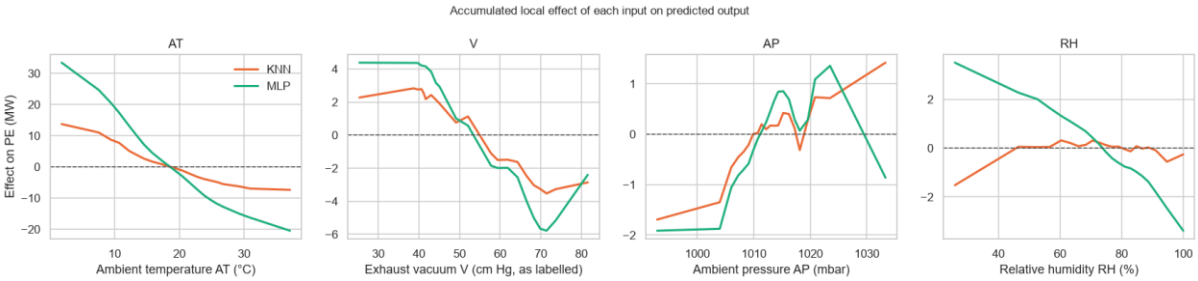


Figure 21. Accumulated local effect of each input on predicted output, MLP and KNN.

Table 11. Learned effects (MLP, accumulated local effects) against the physical expectation.

Input	Expected (section 2.2)	Learned by the MLP	Agrees
AT	strong decrease	decrease, 54 MW over the range	yes
V	decrease	decrease, 10 MW over the range	yes
AP	small increase	small, non-monotonic, 3 MW over the range	yes: small, as expected; no consistent rise, as in the data (section 5.4)
RH	small, sign uncertain (data: decrease at fixed AT)	decrease, 7 MW over the range	yes

- AT: output falls by about 54 MW across the range, in line with the 52 MW gap between the coldest and hottest hours. The curve is steepest between 10 and 25 °C and flattens above 25 °C.
- V: output falls by about 10 MW, almost all of it between about 42 and 72. Above 72 the curve turns slightly upwards; few hours lie there and all are hot, so that end is poorly supported and is not a real gain.
- RH: output falls by about 7 MW from the driest to the most humid hours, close to the 8 MW of section 5.4. The model confirms the sign reversal: at fixed conditions humid air lowers output.
- AP: a small effect of about 3 MW that rises over the middle of the range and falls back at the top, as in section 5.4. The model has learned that pressure matters little here, which matches the data, although ideal-gas reasoning alone would suggest a stronger effect.

KNN's curves have the same directions but are much flatter (21 MW for AT instead of 54), because moving one input a small step rarely changes which six hours are nearest. This does not affect KNN's accuracy on real hours, where all inputs move together.

11.3 The temperature-condenser interaction

If the MLP has learned that V matters mainly in warm weather, its sensitivity to V should depend on AT. For the real training hours in each temperature band, V was raised by 1 unit and the change in predicted output averaged (Figure 22). The sensitivity is largest between 15 and 20 °C (−0.39 MW per unit of V), clear from 10 to 30 °C (−0.17 to −0.19), and near zero below 10 °C and above 30 °C, where V hardly varies. The model has learned the interaction found in section 5.3. At 0.2 to 0.4 MW per unit, a condenser running 10 units worse than normal in spring would cost roughly 2 to 4 MW.

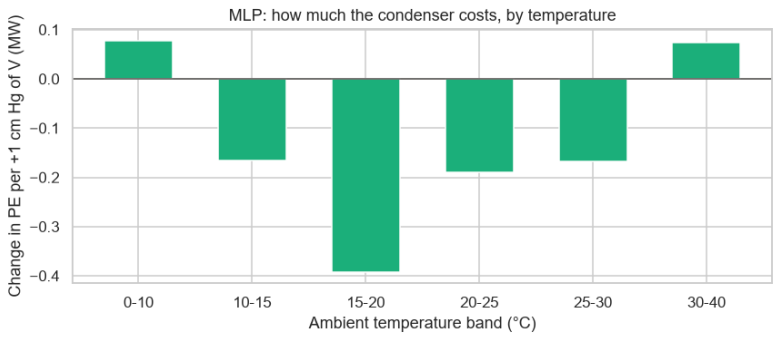


Figure 22. Change in predicted output for a 1-unit rise in V, by temperature band.

11.4 Causes of prediction error

The test RMSE of about 3.5 MW has identifiable sources:

1. Plant states not in the data. The largest errors (24 to 48 MW) are hours where the plant produced far less than the weather allowed, and the three models fail largely on the same hours (15 of the MLP's 24 largest-error hours are shared with both other models). A partial trip or part-load operation within the hour would explain them. They are rare but dominate the worst-case error.
2. Sparse operating conditions. Errors are largest in the extreme output bands, where training hours are fewest, and predictions there are pulled towards the middle.
3. Hourly averaging. Two hours with the same average conditions can contain different short-term variations and so different outputs, which sets a floor on the error of any model of hourly averages.
4. Measurement quality. V is logged at a coarse resolution (634 distinct values) in an unclear unit, and RH exceeds its limit in some hours. Realistic sensor noise alone adds about 0.3 to 0.6 MW to the RMSE (section 9.5).
5. Plant condition over time. Compressor fouling, washing and overhauls change the output for the same weather, and without timestamps or maintenance records the models cannot see them.

11.5 Consequences of wrong predictions and a safety margin

If the plant commits exactly the predicted output, it falls short on about half of all hours, because the errors are centred on zero. Committing a safety margin below the prediction trades shortfall against unsold energy. Figure 23 shows this trade-off from the KNN test errors, scaled to a year of full-load operation (8,760 hours, 3.98 TWh). With no margin, the plant is short on 49.7% of hours; a 2 MW margin cuts that to 23.7%, 5 MW to 5.3% and 10 MW to 0.5%, while unsold energy grows from 10.5 to 87.6 GWh per year.

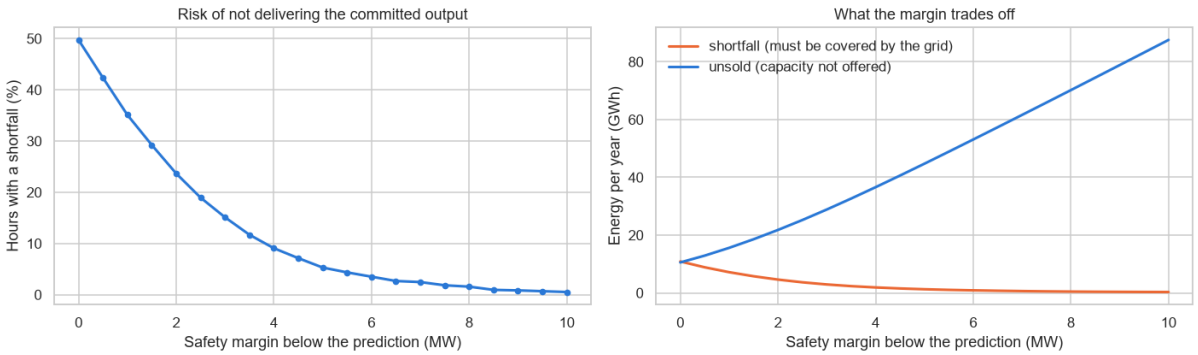


Figure 23. Share of hours short (left) and yearly shortfall and unsold energy (right) as the safety margin grows.

The best margin depends on market prices, which the dataset does not contain, but the structure of the decision is general. If each MWh of shortfall costs c_s and each unsold MWh costs c_u, expected cost is lowest when the margin equals the quantile of the prediction error at level $c_s/(c_s + c_u)$, the classical newsvendor result (Table 12). Because the margin comes from the error distribution, a more accurate

model allows a smaller margin: every MW by which the error band shrinks is a MW that can be sold at the same risk.

Table 12. Cost-optimal safety margin for four penalty ratios, from the KNN test errors.

Shortfall penalty / sale price	Quantile level	Optimal margin (MW)	Hours short (%)
1	0.50	−0.03	50.0
2	0.67	1.15	33.3
4	0.80	2.41	20.0
9	0.90	3.79	10.0

12. Limitations

- One plant, one operating mode. The models describe this plant at full load and do not transfer to another plant or to part load without new data.
- No time information. Without timestamps the split cannot be chronological and drift over the six years cannot be modelled. If neighbouring hours are similar, a random split can make the test score slightly optimistic.
- Unrecorded plant events. Trips, partial outages and maintenance are not marked, so the largest errors cannot be explained or prevented with these inputs.
- Measurement issues. The unit of V contradicts its label and RH exceeds 100% in some hours. The models are unaffected, but V cannot be read as a true pressure.
- Range of validity. All models are unreliable outside the observed conditions. The envelope and density checks flag such hours but cannot correct them.
- Equal weighting in KNN. The selected model cannot isolate single-input effects (section 10.1), so the MLP is used for interpretation.
- Comparison with the reference study. The reference used a different split and does not report removing the duplicates, so the comparison is indicative.

13. Recommendations

1. Deploy the KNN model with both reliability checks for hourly performance monitoring. Raise an alarm when the measured output stays more than about 5 MW below the prediction for several consecutive hours, and show the six reference hours with each alarm.
2. Commit to the grid with a margin, not the bare prediction. Choose it from the error distribution and the actual penalty ratio (section 11.5), and recompute it whenever the model is retrained.
3. Use the three-input model (AT, AP, RH) for day-ahead commitments, since V is not known in advance. It costs under 1 MW of RMSE.
4. Schedule condenser cleaning and cooling-system checks before the warm season. The condenser penalty is largest between 15 and 30 °C.
5. Evaluate inlet-air cooling with care. Temperature is the dominant loss (about 1.5 to 2.2 MW per °C), so cooling the compressor inlet on hot afternoons is the largest lever. Evaporative coolers

also raise humidity, which the models show lowers output, so part of the gain is given back. The models can estimate the net effect of a proposed cooler within the observed conditions.

6. Fix the data at the source: calibrate the RH sensor, confirm the unit and measuring point of V, record timestamps, and log trips, part-load hours and maintenance. With timestamps the model can be validated chronologically and used to track degradation.
7. Retrain regularly and watch the residuals. A steady drift of measured output below prediction over weeks is the signature of gradual fouling.

14. Conclusions

1. The net output of the plant at full load can be predicted from four plant measurements to a test RMSE of 3.45 MW and MAPE of 0.540% (R^2 0.959) with KNN regression, well inside the 1% target and better than the reference study's best result (R^2 0.912, MAPE 0.872%). SVR and MLP also beat the reference.
2. The audit found problems the reference study did not report: 41 duplicate records, 55 humidity readings above 100%, and an exhaust vacuum unit that cannot be correct. Handling them with fixed, logged rules removed a source of leakage without hiding any real operating condition.
3. KNN was chosen on evidence beyond accuracy: the fewest costly over-predictions, the best accuracy at the extremes, and predictions backed by real past hours. Tests beyond the data showed that no model can be trusted outside its training range, and that the SVR fails dangerously in a heatwave, so the deployed model carries range and density checks.
4. What the models learned agrees with thermodynamics: output falls by about 54 MW across the temperature range, the condenser costs output mainly between 15 and 30 °C, humid air lowers output at a fixed temperature, and pressure matters little at this site.
5. The model supports two plant decisions: a performance alarm for condition monitoring and a cost-based safety margin for grid commitment. Both are available interactively in the Tableau Public dashboard.

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[7] Apley, D. W., & Zhu, J. (2020). Visualizing the effects of predictor variables in black box supervised learning models. *Journal of the Royal Statistical Society: Series B*, 82(4), 1059-1086.

Appendix A. Repository contents and reproduction

Table 13. Contents of the repository.

Folder	Contents
01_Dataset/raw	Folds5x2_pp.xlsx (original UCI file, unchanged) and raw_data.csv (Sheet1 as CSV, unchanged)
01_Dataset/cleaned	cleaned_data.csv (9,527 rows), cleaning_log.csv, cleaning_changes.csv (96 row-level changes)
02_Notebook	CCPP_Power_Prediction.ipynb, the full analysis; outputs/ holds every figure and CSV it writes
03_Dashboard	CCPP_dashboard.twbx (published workbook), CCPP_dashboard_walkthrough.mp4 (2:49), screenshots/, data/ (seven CSVs)
04_Report	This report (PDF)
05_Presentation_and_Poster	Slides and poster (PDF)

To reproduce every result, open *02_Notebook/CCPP_Power_Prediction.ipynb* in Google Colab, choose Runtime > Run all and upload *Folds5x2_pp.xlsx* when asked; run locally from inside *02_Notebook/*, the notebook finds the data without an upload. All random processes use seed 42, so a second run gives the same numbers. A full run takes about 10 minutes on an 8-core laptop and longer on free Colab, mostly for tuning the SVR and MLP. The software versions used were Python 3.14, NumPy 2.5, pandas 3.0, scikit-learn 1.9, Matplotlib 3.11 and seaborn 0.13.